

Addressing Challenges and Opportunities in Online Extremism Research: An Interdisciplinary Perspective

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ABSTRACT

Recent political and cultural upheavals around the globe highlight the importance of studying extremism in online spaces. The CSCW community has shown growing interest in exploring the information and influence operations, use of language and multimedia and finance structures in online extremism movements. The goal of this workshop is to broaden the current research by considering multi-disciplinary and cross-cultural aspects of online extremism research. This workshop brings together international researchers to discuss challenges and new opportunities to help (1) understand the representativeness and validity of the data used for studying extremism (2) uncover nuances in extremism research across countries and cultures (3) build norms for platform governance and policies (4) develop critical perspectives on extremism studies addressing questions such as research ethics and bias (5) promoting research efforts to aid victims of the extremism (6) leverage extremism research for aiding the recovery and social support for populations involved in online extremism.

CCS CONCEPTS

• **Human-centered computing** → Social media.

KEYWORDS

online extremism, data, content moderation, culture, policy, research ethics, collaboration

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1 INTRODUCTION

Extremist movements target unsuspecting audiences to promote their fundamentalist ideologies and agendas [16, 23]. Their goals of vilifying people for inherent characteristics such as race, nationality, religion, gender identity, sexual identity, and political views often result in the recruitment of new members who support the extremist cause [2]. Proponents of extremist ideologies increasingly utilize online platforms to disseminate mis- and disinformation, call for donations, and send invitations to alternative platforms with low moderation and links to podcasts and shows discussing extremist ideologies [26]. Scholars perceive this growing ecosystem of biased and polarized information as a threat to digital democracy and a gateway to radical collective action [32].

In recent times, research on online extremism has increased substantially; however, there are several under-explored research topics that are crucial for extracting practical and actionable insights from the scholarly research. This workshop builds on the literature discussing following challenges in extremism studies that we identified based on the prior literature and the organizers' research experiences: 1) Identifying and Validating Data for Research, 2) Incorporating Cultural Context in Extremism Research, 3) Establishing Norms for Platform Governance Policies for Extremism, 4) Establishing Extremism Research Ethics 5) Balancing Focus on the Victims and Perpetrators of Online Extremism 6) Facilitating Recovery from Online Extremism.

In this workshop, we aim to reach beyond the concepts of online incivility, toxicity, hate speech, harassment or online harm and also consider ideologically driven extremism based on race, ethnicity, gender, sexual orientation, religion, nationality, caste, and political affiliation. As such, we invite ideas researching both, online harms, hate speech and harassment relevant to ideological extremism and broader level online ecosystem of ideological extremism. Keeping in mind the socio-political complexity of ideological extremism, this workshop focuses on bringing together researchers from diverse countries and disciplines such as computational social science, HCI, criminology, social psychology, epidemiology and political science to facilitate further dialogue on six main themes of the workshop.

2 WORKSHOP THEMES

2.1 Identifying and Validating Data for Online Extremism Research

Various extremist groups have become adept at leveraging emerging online technologies [1, 5, 6] and the participatory culture of online social media [27, 32]. Given the massive amount of information produced online, online extremism researchers are confronted with almost infinite - but non-validated - data generated in real time [13]. However, only a handful of studies exploring online extremism include a discussion on challenges faced in the data collection and validation processes. For example, Crosset et. al mention challenges faced while identifying far-right accounts in Canada arising due to the lack of consensual definitions of various extremist ideologies [11]. For instance, there is no theoretical consensus on the defining properties of the far-right ideology in the United States [7]. In fact, one study found that the majority of respondents in the United States do not have a clear conceptualization of the ideological left and right, and thus the left-right placement is a more symbolic than substantive measure [22]. This poses a challenge in identifying conceptual boundaries of extremist ideologies in research design.

Another challenge in identifying extremist content online is lack of clearly labeled extremism related data or accounts relevant to online extremism [13, 14]. First, while some accounts openly declare allegiance to known extremist groups, many relevant accounts operate with pseudonyms with no self-disclosed relation to any extremist ideology [7]. Second, the extremely fluid nature of online conversations may result in the deliberate dissociation of the content from its original source, a practice commonly referred to as decontextualization [12]. Decontextualization makes the process of identifying the sources of extremism considerably more complicated [11].

These are but a few challenges in studying the increasingly relevant problem of online extremism. To the computational social science scholars, we ask: How do we derive novel data collection and validation methods for facilitating research in online extremism? To the theoretical researchers in CSCW we ask: What is our role in contributing and solidifying constructs of various extremist ideologies in the digital context?

2.2 Incorporating Cultural Context in Extremism Research

Extremist and hate groups are profiting from unprecedented growth in online audiences across the world [19, 31]. However, the majority of online extremism studies focus on the USA and Europe [10, 21]. Considering cultural and national context could be crucial towards identifying extremist discourse and ideological constructs across cultures. For example, some types of extremism are strongly related to the country or the region (e.g., Irish Republicanism, Ulster Loyalism), some related to religion (e.g. Islamic terrorism in multiple contexts, like ISIS, Boko Haram, etc), culture (e.g. neo-Nazism), or ideology (e.g. far right in the US, India) [9]. Moreover, expressions of extremism could be strongly conditioned by culture-driven, emotional experiences [24].

The context of the country or culture can further inform the representativeness of extremism studies. For example, while right-wing extremism has greatly outpaced left-wing and other types of extremism in the USA [18], other countries such as Germany have more varied forms of extremism [20, 29]. Indeed, the group or groups who are primarily targeted by online extremism varies cross-national [30].

In light of these insights, we invite CSCW scholars to reflect upon the cross-cultural aspects of studying online extremism. Specifically, we ask ethnographers, social political and computational scientists: How can we systematically incorporate linguistic, cultural and political context of the researched populations in the online extremism studies?

2.3 Establishing Norms for Platform Governance Policies for Extremism

Is online extremism ungovernable? Ganesh et. al. attribute the ungovernability of digital hate communities to three characteristics: its decentralized and amorphous structure, ability to quickly migrate communities across platforms and use of coded language to subvert content moderation [15]. In addition to the difficulties in governing extremist communities, there is also a larger debate on exactly which type of content from such communities needs moderation. For example, while explicit threats of violence, explicit calls for violence, privacy invasions, and technological attacks are commonly moderated, discriminating speech directed at minorities is overlooked because it contains online expressions, not actions [8]. However, researchers argue that online hate can help frame violence as something acceptable when directed at specific groups in society [8]. Specifically, online expressions of hate can incite the readers to cause violence, interfere in social relationships in online networks and cause emotional distress in the primary and secondary targets of hate [3]. Perhaps due to the lack of ability to distinguish between different types of extremism and types of harms, online platforms may be cherry-picking cases of online extremism to moderate [25, 26].

Hence, we call upon the expertise of the digital policy and content moderation researchers to reflect on how we can contribute to bettering the norms of moderating extremism online. Specifically we ask: what are the practical challenges and roadblocks in moderating extremism online? How do we design content moderation systems to reduce platform migration of extremist communities?

2.4 Establishing Extremism Research Ethics

Research ethics in studying online extremism is a largely unexplored topic within the CSCW community. Especially with respect to big data studies in extremism, it is not surprising that large scale models could be built to surveil individuals related to extremist activities [4]. Even though such models are built with intentions to minimize the social harm caused by extremism, we need to ask: to what end such large scale models will be used and who has the data and access to modify them?

Moreover, it also needs to be debated whether similar ethical standards apply when researching extremism compared to other positive social justice movements. In all, online extremism research poses numerous ethical dilemmas which need to consider social

justice, digital democracy, privacy and freedom of speech. This workshop aims to involve researchers from multiple disciplines such as criminology, social science, social justice, epidemiology, HCI ethics into advancing the dialogue towards setting actionable norms for studying and modeling extremism online. To the CSCW community we ask: How can we design norms for data sharing and modeling in online extremism research?

2.5 Balancing Focus on the Victims

Resources for research, including time and funding, are not infinite. In the context of extremism research, it is possible that work focused on perpetrators will gradually contribute to a reduction in violence. However, it may be that increasing the focus on extremist perpetrators will inevitably detract from efforts to support their victims or defend their potential victims. For example, research into white supremacy can be critiqued for centering whiteness and white experiences, or for taking up funding opportunities that might otherwise go to supporting those who suffer in the context of a white supremacist system.

In light of this issue, we invite scholars to reflect upon whether there are ethical or methodological guidelines to help shape research into extremism.

2.6 Facilitating Recovery from Extremism

Finally, the goal of this workshop is to also introduce a novel problem to the CSCW community: recovery from online extremism. Social psychology researchers have highlighted the role of social exclusion and ostracism in engagement with extremist ideologies [17]. Similarly in online setting, heavy ostracism from other platform users was found to be a precursor of joining conspiracy discussion communities [28]. While social exclusion and ostracism lead to engagement with extreme ideologies, can social support and acceptance help recovery from extremism movements?

Reddit communities such as *r/ReQovery*, *r/RecoveringIncels* and *r/exredpill* designed to host users recovering from fringe movements such as QAnon, Incels, Redpill open up promising direction of research that could explore the role of technology in aiding the recovery from extremism. We invite the CSCW and HCI community to discuss how technology and platform design can help users disengage from online extremist communities or provide forums for sharing their experiences. Specifically, to the HCI researchers we ask: How can we design platform or content interventions to aid disengagement from the extremist communities? How can we shape online spaces to better support recovery from extremism?

3 WORKSHOP ACTIVITIES AND DETAILS

3.1 Activities

The workshop will start with a around-the-room introduction of the participants. We will then invite a guest speaker with expertise in extremism and social justice for a keynote address. This will be followed with workshop paper presentations and Q&A. In the second half of the workshop, we will hold a round table, moderated discussion on the five main themes of the workshop. Finally, we also want to create an opportunity for researchers working in this space to share their research experiences that are often emotionally taxing. To facilitate this, we will make use of the virtual setting and create

breakout rooms for small group discussions where participants might more comfortably discuss their experiences and future goals. We anticipate that the participants will only require access to the virtual conference and zoom room for the workshop.

3.2 Participants

We expect a maximum of 50 participants for this workshop. We will primarily use social media to recruit participants from various disciplines in extremism research. Our multidisciplinary team of organizers will allow us to recruit from a diverse scholarly audience that can bring valuable perspectives to the workshop. We also plan to leverage CSCW's international reach to encourage participants from various countries.

3.3 Workshop Goals

In summary, we have following goals for the workshop:

- (1) Facilitate networking, connections and community identity of researchers from multiple disciplines such as social psychology, computer science, criminology, epidemiology, social justice studying online extremism.
- (2) Discuss and propose ideas relevant to the five themes of the workshop.
- (3) Encourage future interdisciplinary collaborations on the ideas generated during the workshop

4 ORGANIZER BACKGROUND

Shruti Phadke

Shruti Phadke is a PhD student at the University of Washington Information School. Her research focuses on studying the social dynamics and linguistic expressions in online fringe communities (eg. domestic extremist groups, conspiracy theorists). She has a publication history in CSCW and CHI with a Best Paper Honorable Mention award for the most recent CSCW paper. Her research methods include quantitative and mixed methods analysis of large scale social media data in context of theories from social psychology and sociology. She has previously received her Masters from Virginia Tech and has worked as a research intern at UT Austin Psychology Lab, Oaks Ridge National Laboratory and Kitware.

Jessie Seiler

Jessie Seiler is a second year doctoral student in the Epidemiology department at the University of Washington's School of Public Health. Jessie has worked in behavior change communications with domestic and global health non-profits and the Centers for Diseases Control and Prevention. She earned her MPH from the Rollins School of Public Health at Emory University, where her research interests coalesced around race-based violence. Her current focus is on the online and in-person epidemiology of white supremacy.

Tanushree Mitra

Tanushree Mitra is an Assistant Professor at the University of Washington Information School, where she leads the Social Computing research group. She and her students study and build large-scale social computing systems to understand and counter problematic information online. Her research spans auditing online systems for misinformation and conspiratorial content, understanding digital misinformation in the context of the news ecosystem, unraveling

narratives of online extremism and hate, and building technology to foster critical thinking online. Her work employs a range of interdisciplinary methods from the fields of HCI, data mining, machine learning, and natural language processing. Tanu's work has been supported by grants from the NSF, Social Science One, and DoD.

Kiran Garimella

Kiran Garimella is the Michael Hammer postdoc at the Institute for Data, Systems and Society at MIT and an incoming faculty at Rutgers School of Communication and Information. Before joining MIT, he was a postdoc at EPFL, Switzerland. His research focuses on using digital data for social good, including areas like polarization, misinformation and human migration. His work on studying and mitigating polarization on social media won the best paper awards at WSDM 2017 and WebScience 2017. Kiran received his PhD in computer science at Aalto University, Finland, and Masters & Bachelors from IIIT Hyderabad, India. Prior to his PhD, he worked as a Research Engineer at Yahoo Research, Barcelona, and QCRI, Doha.

Matthew Costello

Matthew Costello is an assistant professor of sociology and criminal justice at Clemson University. His research focuses on online hate and extremism, and domestic and cross-national political violence and rebellion. His recent work has been funded by the National Institute of Justice, and he is currently conducting research funded by the National Science Foundation that investigates online extremist behavior related to the COVID-19 pandemic.

James Hawdon

James Hawdon is a professor of sociology and Director of the Center for Peace Studies and Violence Prevention at Virginia Tech. His research focuses on how communities influence the causes and consequences of violence. He is currently researching how online communities influence online hate, extremism, political polarization, and cyber crime. His recent work has been funded by the National Institute of Justice and the National Science Foundation, and he serves on the advisory board for the European Union's Network of Excellence in Training on HATE' (NETHATE) project.

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